Project Output Summary

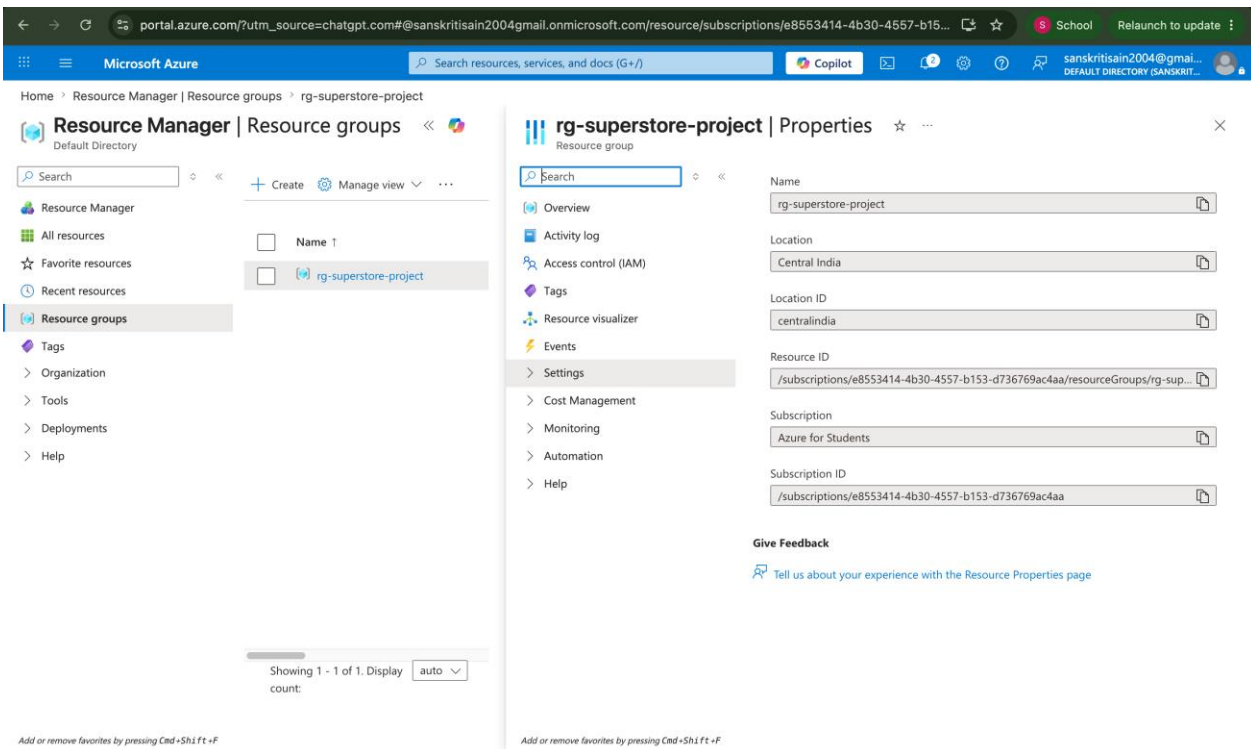
Project Title

Azure Cloud Fundamentals and Data Pipeline Implementation using Azure Data Factory

Screenshots Collected

1. Resource Group

Screenshot: 01_resource_group.png

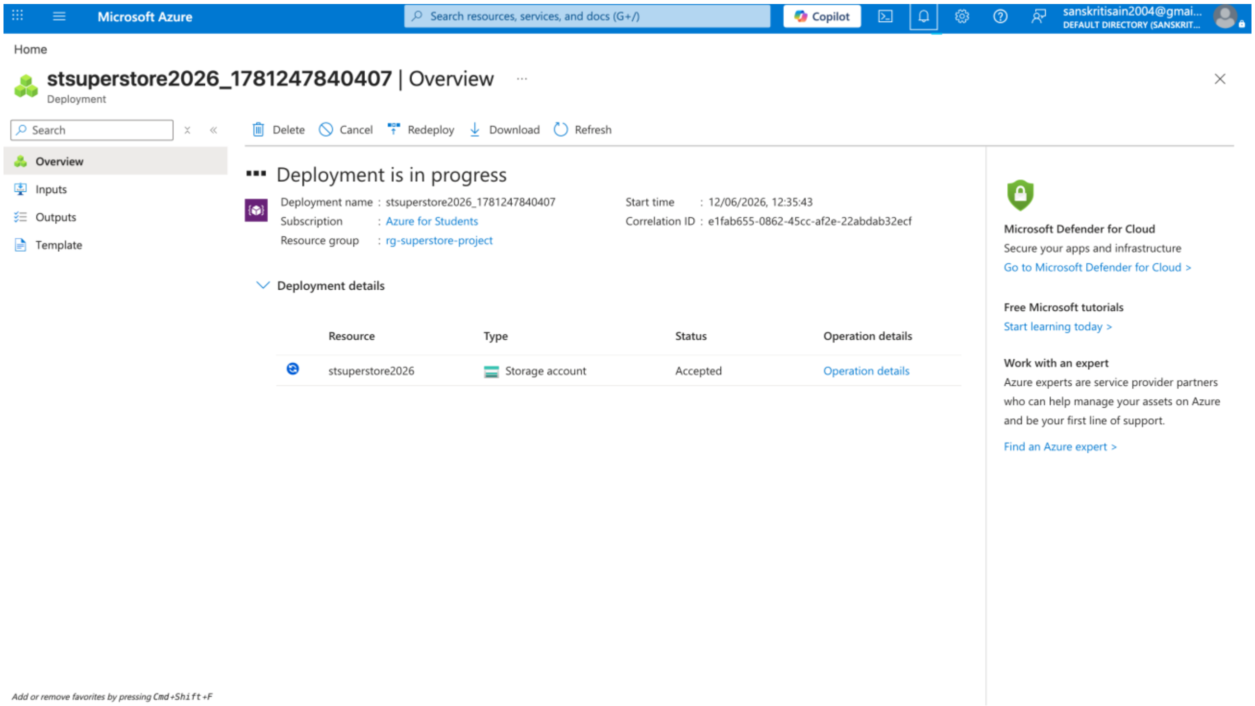


Description:

A Resource Group named rg-superstore-project was created to organize and manage all Azure resources used in this project.

2. Storage Account

Screenshot: 02_storage_account.png

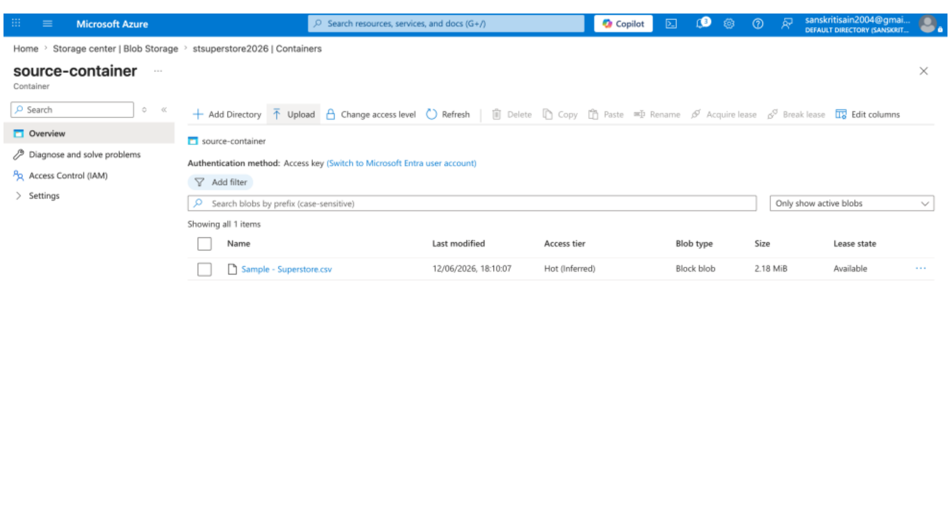


Description:

An Azure Storage Account was created to store source and destination data files used in the pipeline.

3. Blob Containers and CSV Upload

Screenshot: 03_blob_container.png



Description:

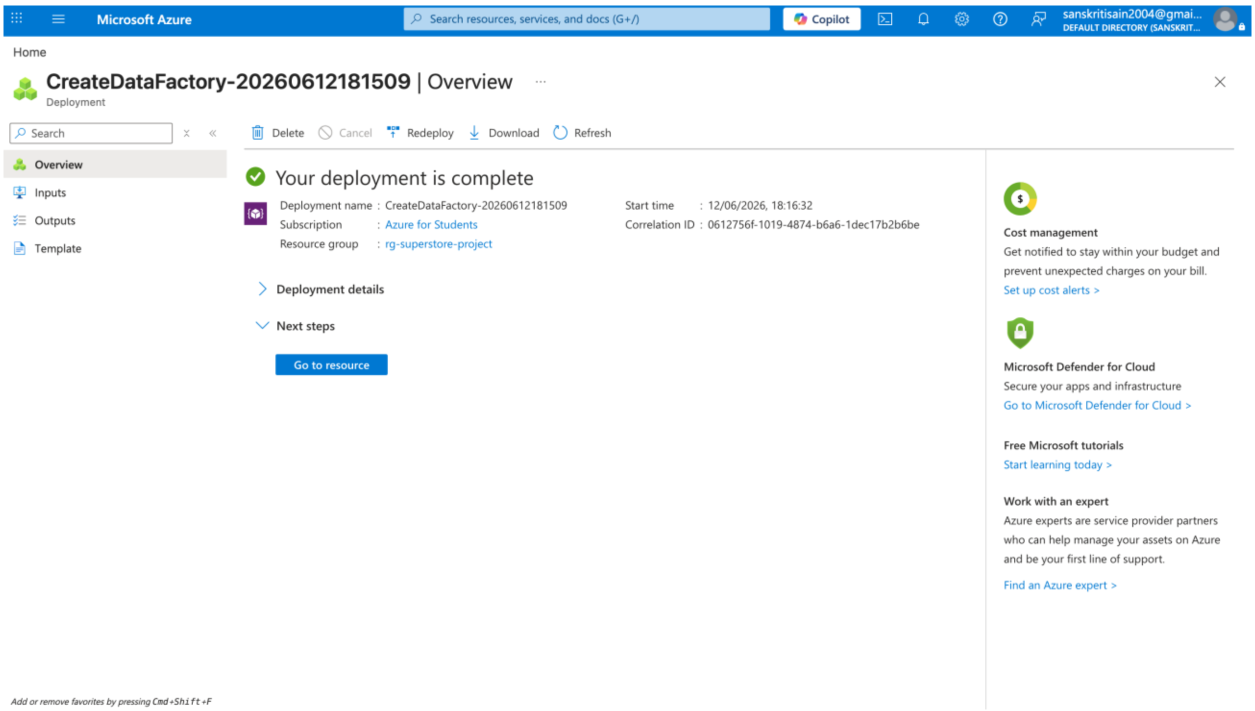
Two Blob Storage containers were created:

- source-container
- destination-container

The file Sample-Superstore.csv was uploaded successfully to the source-container.

4.Azure Data Factory Created

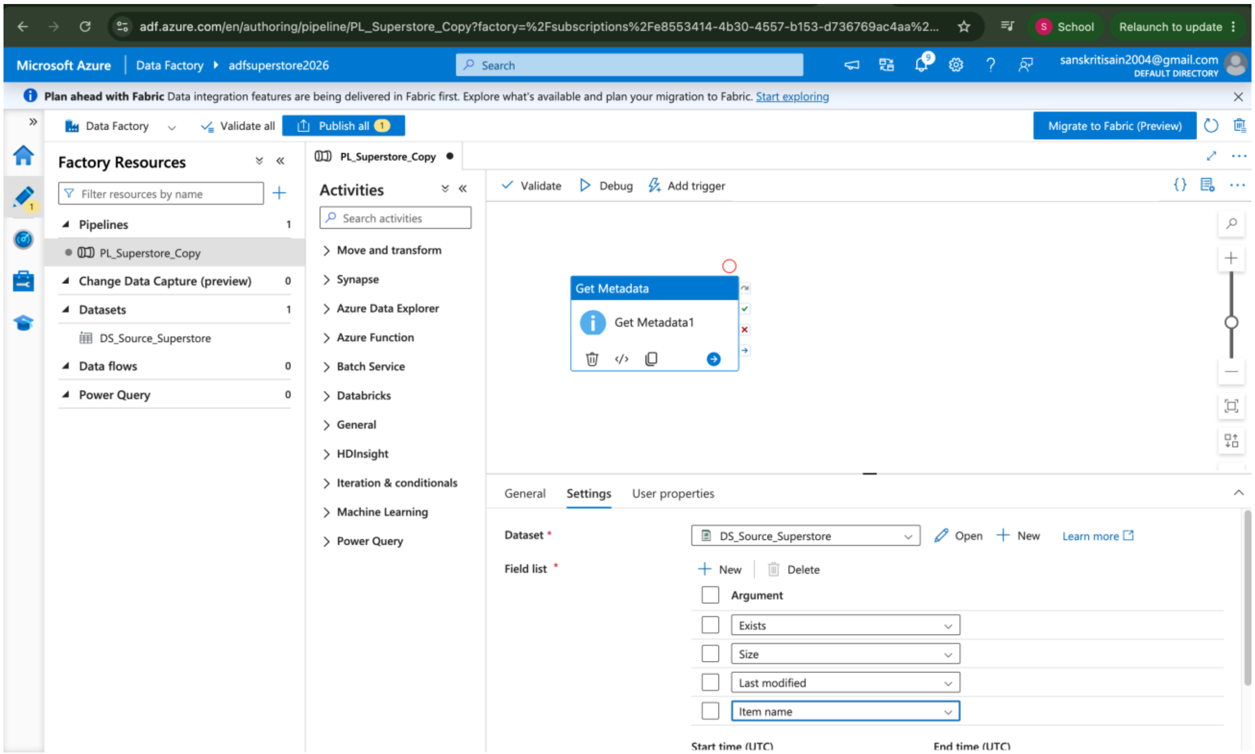
Screenshot 4: 04_azure_data_factory_created.png



Description:
An Azure Data Factory instance was successfully created to build and manage the data pipeline.

5. Azure Data Factory

Screenshot: 05_adf_ui.png

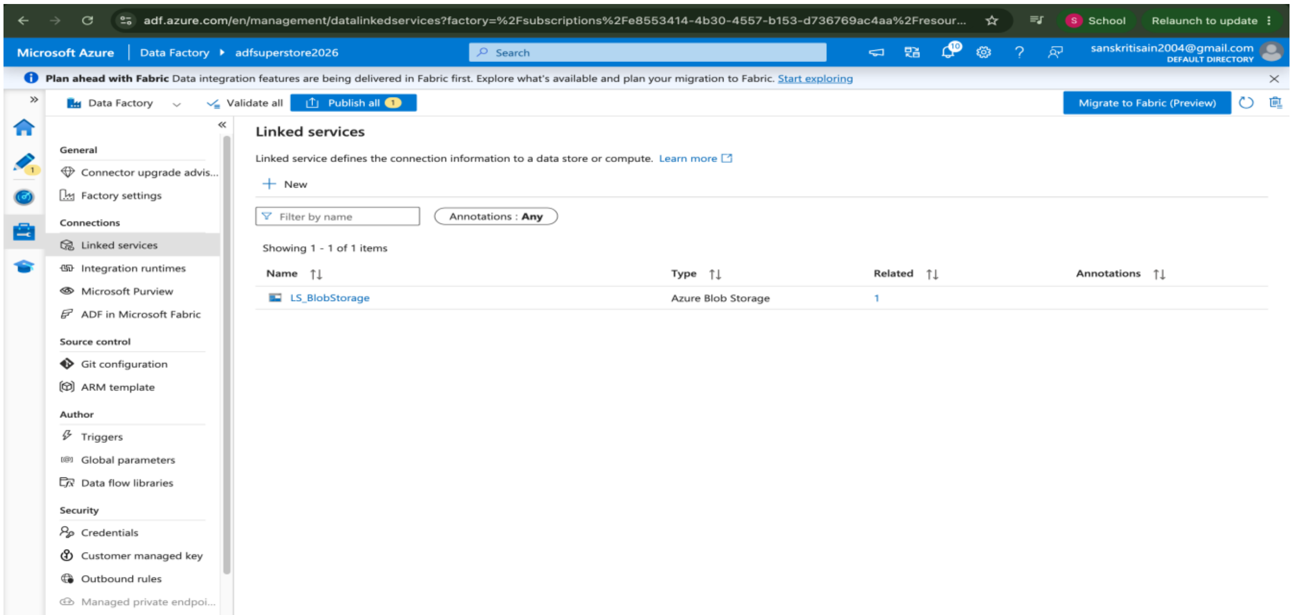


Description:

Azure Data Factory was created and explored using the Author, Monitor, and Manage modules.

6. Linked Service

Screenshot: 06_linked_service.png

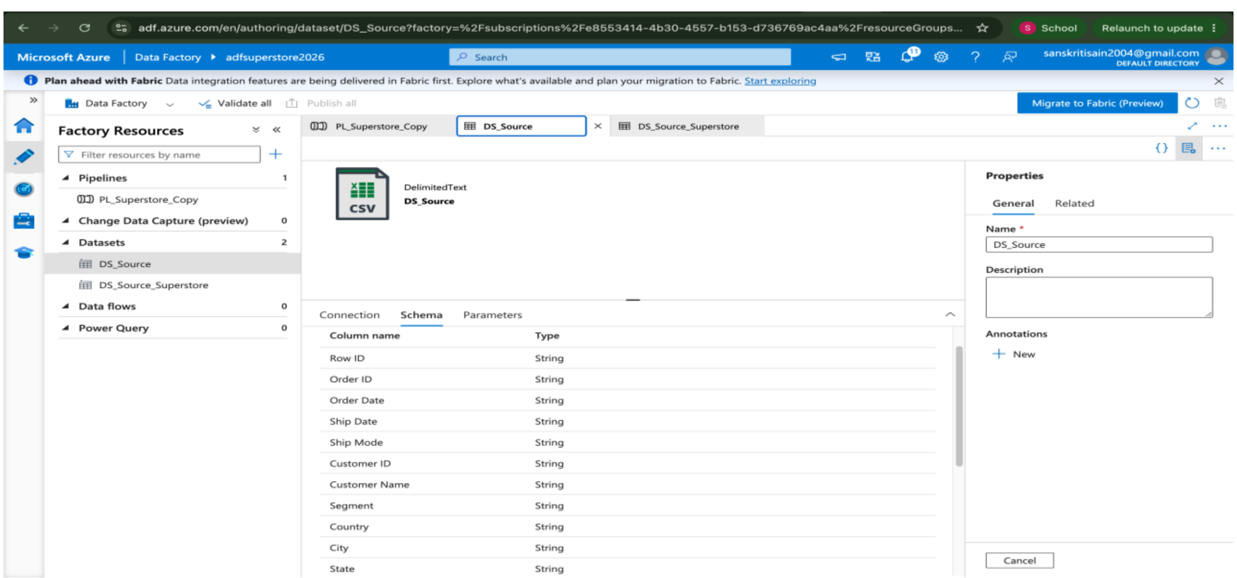


Description:

A Linked Service was created to establish a connection between Azure Data Factory and Azure Blob Storage.

7. Source Dataset

Screenshot: 07_source_dataset.png

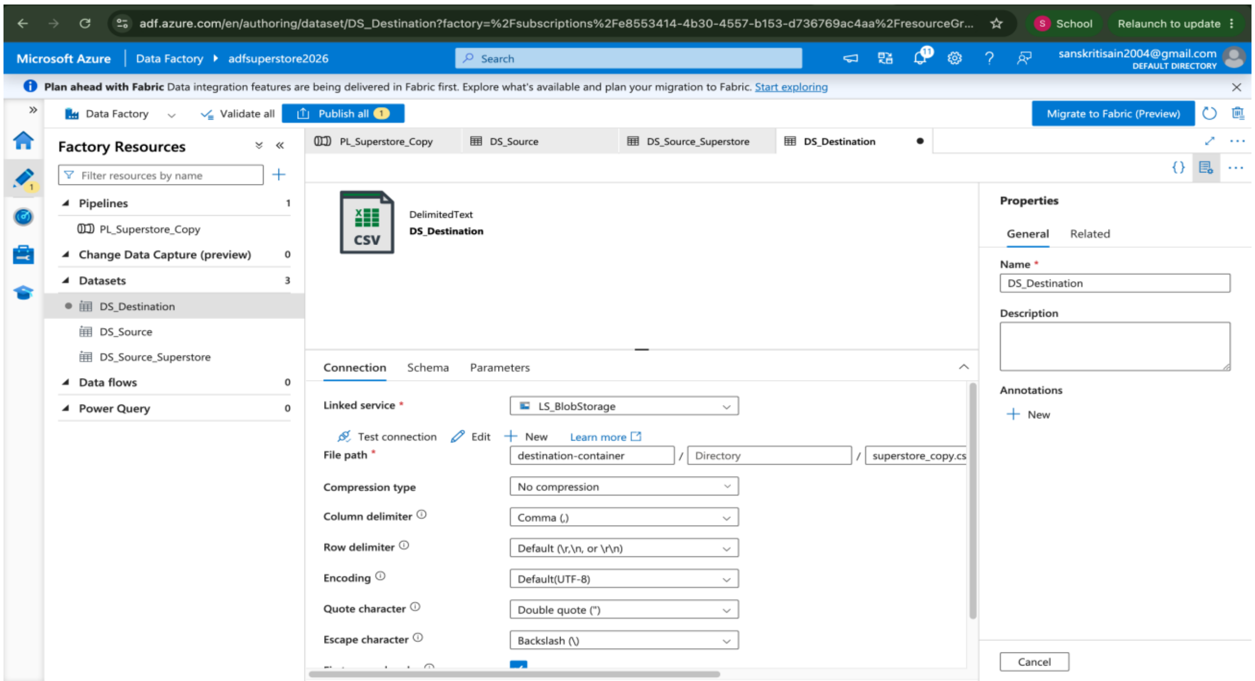


Description:

The source dataset was configured to read data from the file Sample-Superstore.csv stored in the source-container.

8. Destination Dataset

Screenshot: 08_destination_dataset.png

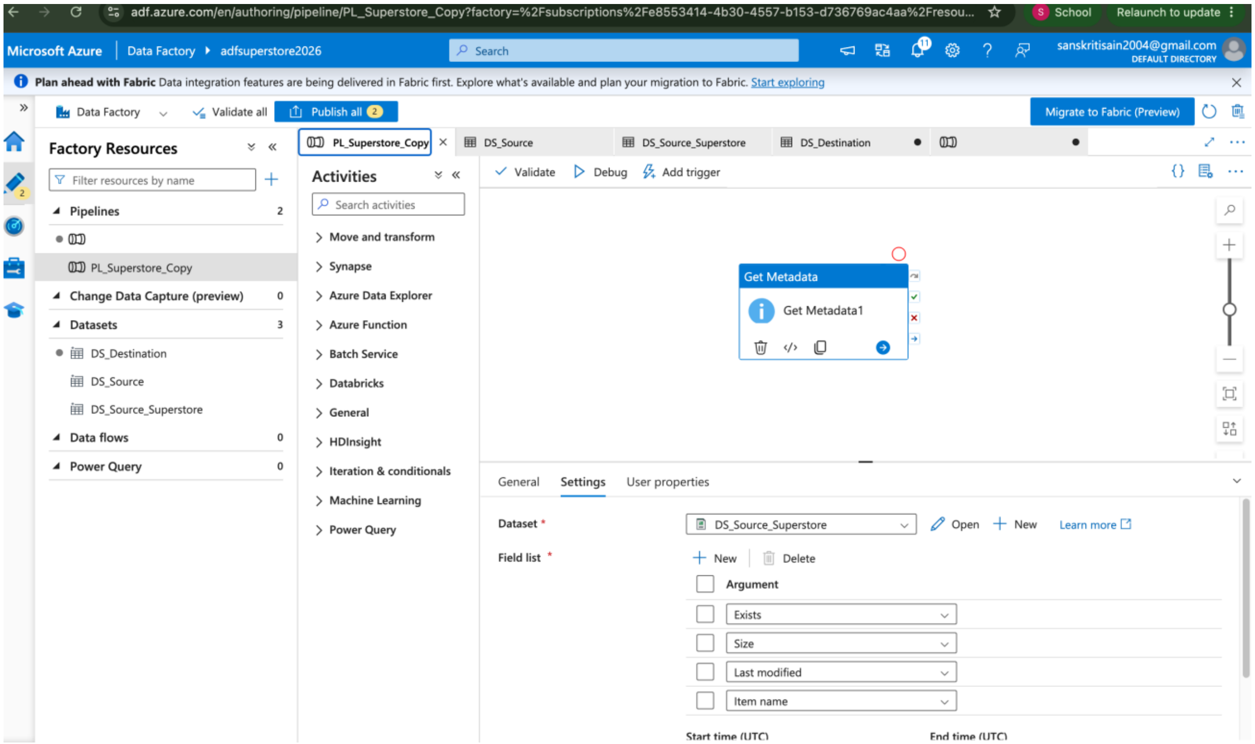


Description:

The destination dataset was configured to write data to superstore_copy.csv in the destination-container.

9. Get Metadata Activity

Screenshot: 09_get_metadata.png

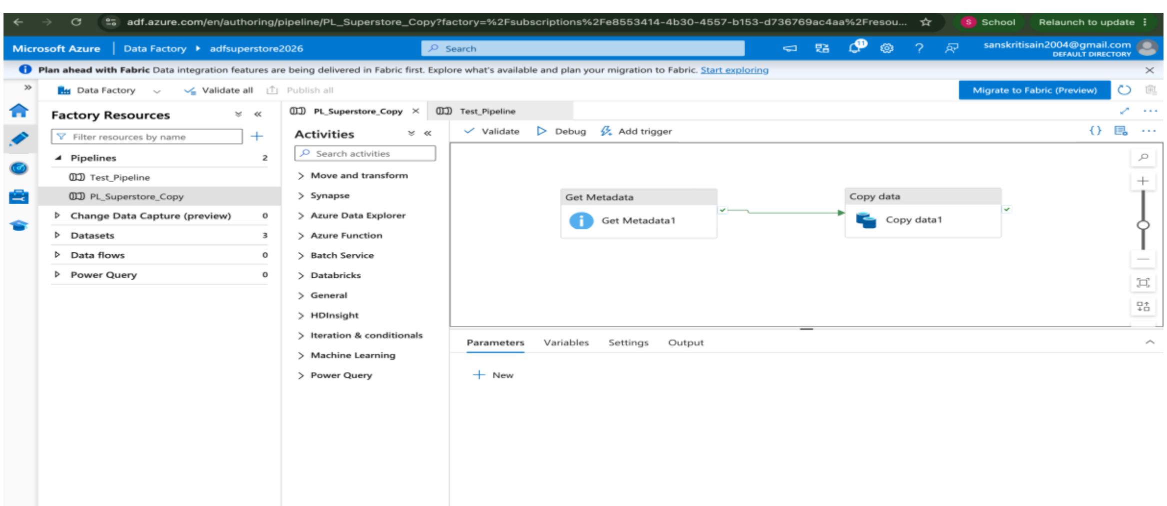


Description:

The Get Metadata activity was configured to validate the source file before processing. The activity checks file existence and metadata information.

10. Pipeline Design

Screenshot: 10_pipeline_design.png



Description:

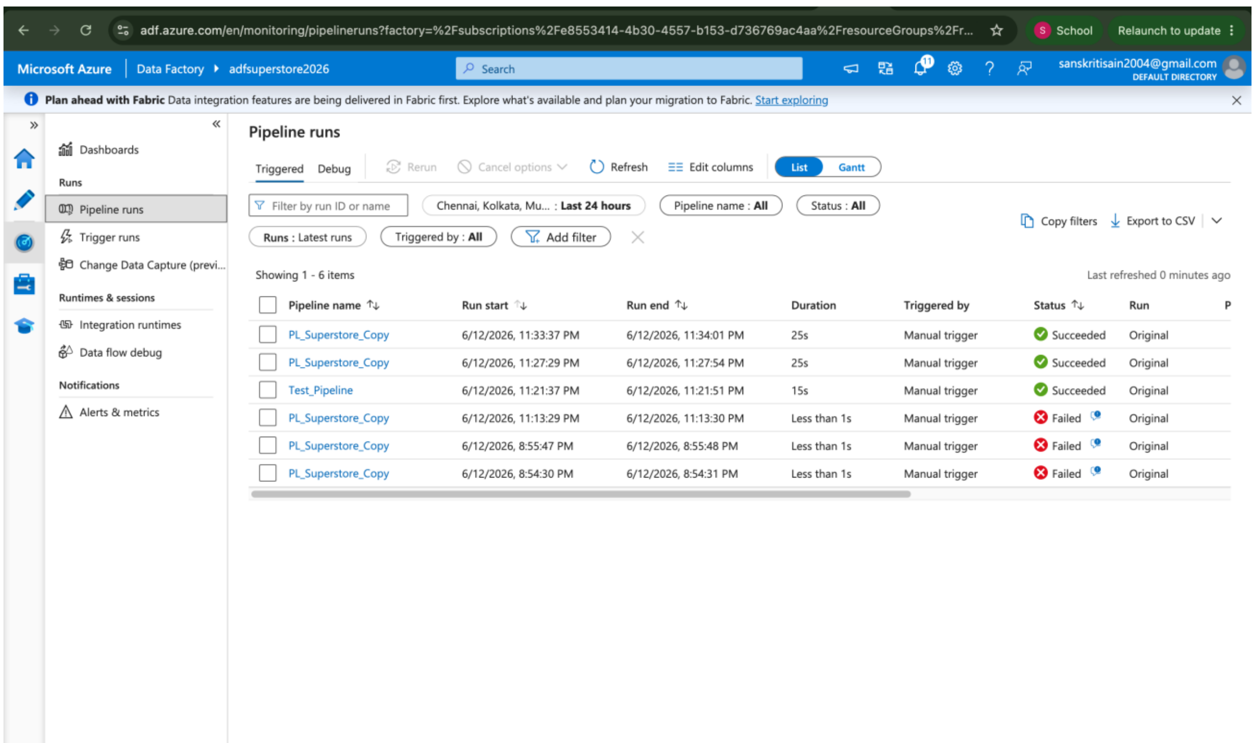
A pipeline named PL_Superstore_Copy was created using:

- Get Metadata Activity
- Copy Data Activity

The activities were connected to form a complete workflow.

11. Pipeline Execution

Screenshot: 11_pipeline_success.png



Description:

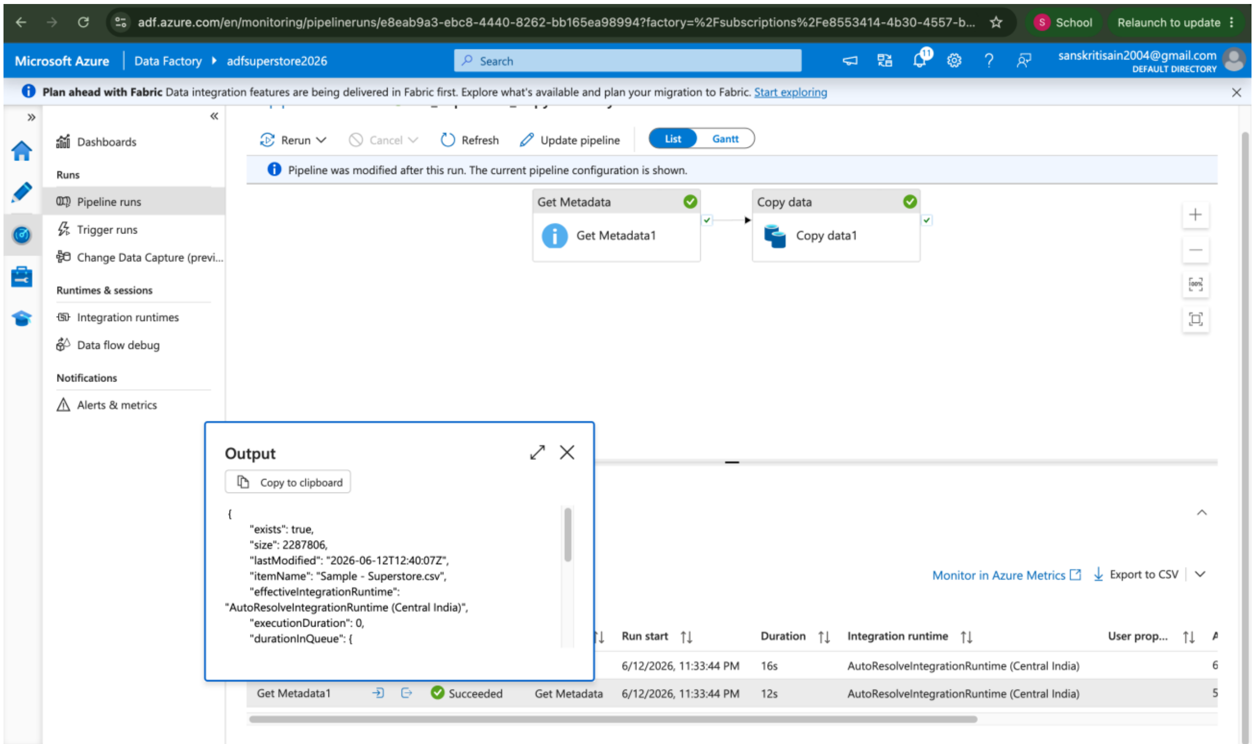
The pipeline was executed successfully using Azure Data Factory Monitor.

Pipeline Status:

- Get Metadata → Succeeded
- Copy Data → Succeeded

12. Metadata Validation

Screenshot: 12_metadata_validation.png

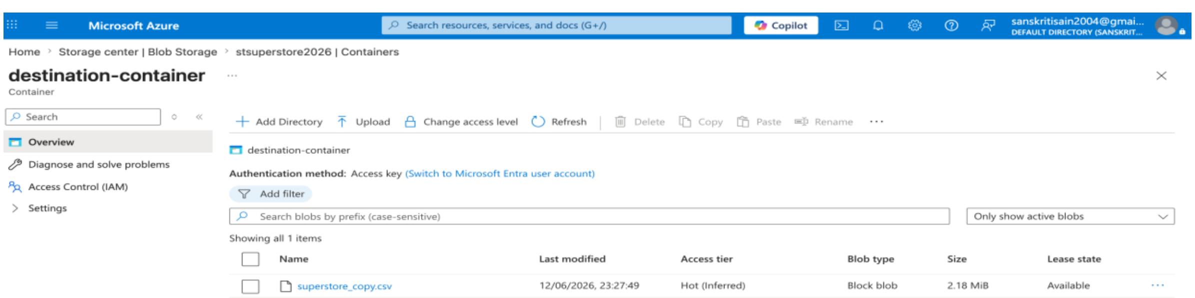


Description:

Metadata validation was completed successfully using the Get Metadata activity before the file copy process.

13. Destination File Created

Screenshot: 13_destination_file_created.png

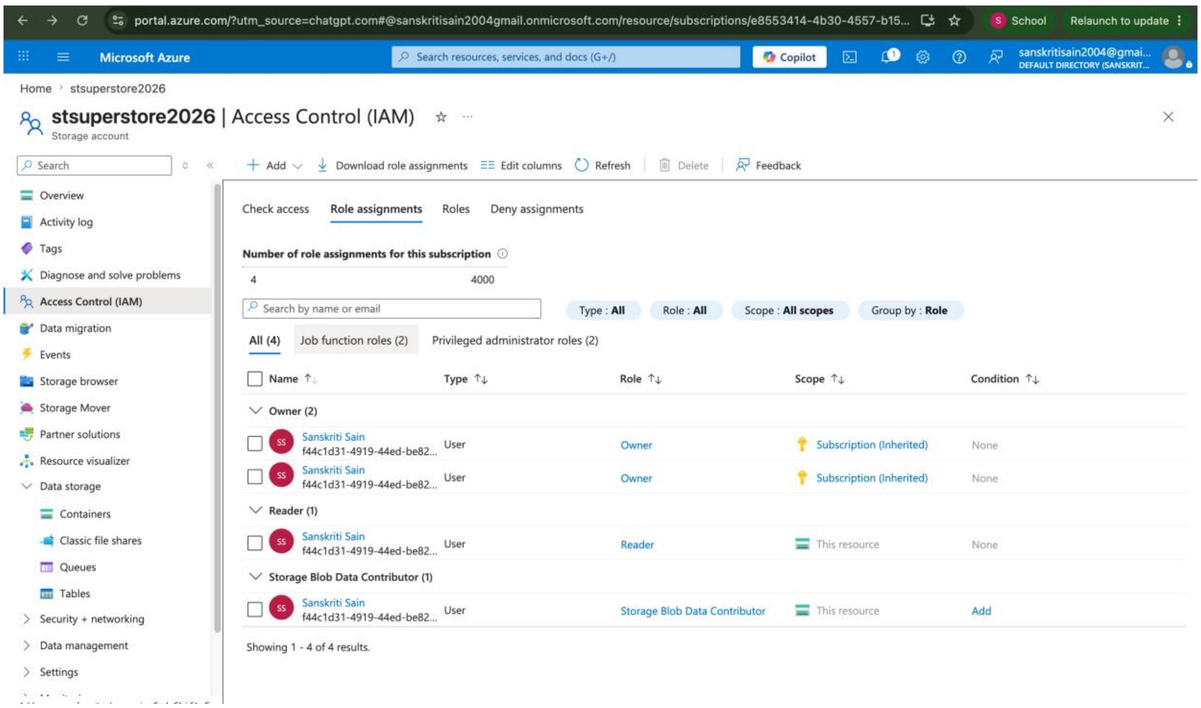


Description:

The file superstore_copy.csv was successfully generated in the destination-container after pipeline execution.

14. IAM Role Assignment

Screenshot: 14_role_assignment.png



Description:

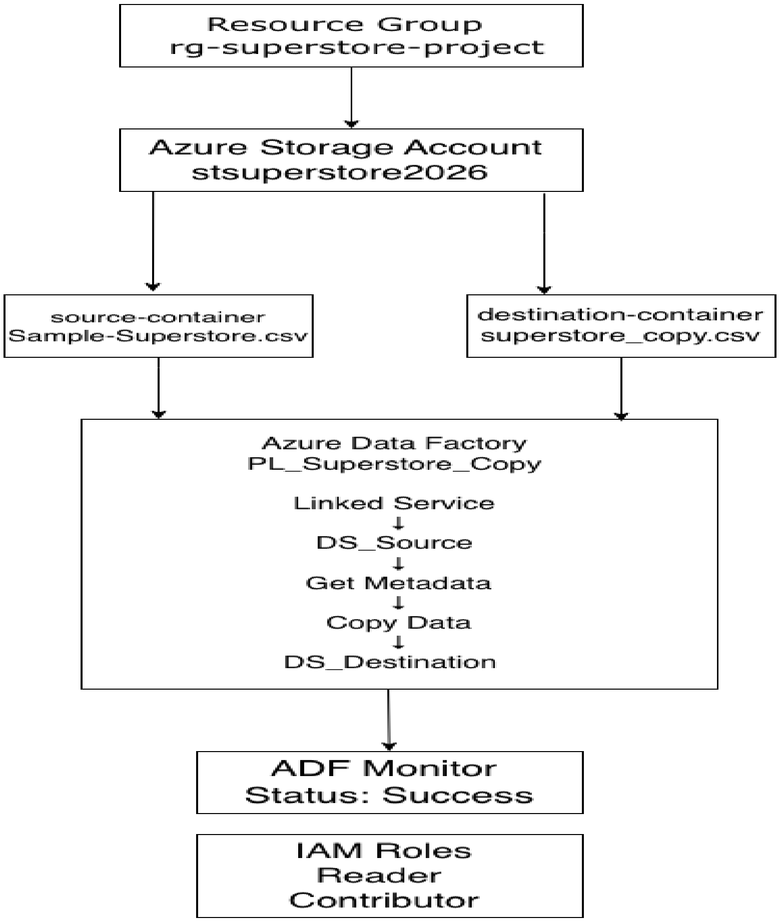
IAM permissions were configured to provide secure access to Azure resources.

Assigned Roles:

- Reader
- Contributor / Storage Blob Data Contributor

15. Architecture Diagram

Screenshot: 15_architecture_diagram.png



Description:

The architecture diagram illustrates the complete flow from Blob Storage to Azure Data Factory and finally to the destination container.

Pipeline Execution Results

Activity	Status
Get Metadata	Succeeded
Copy Data	Succeeded
Pipeline Execution	Succeeded
Metadata Validation	Successful
Data Transfer	Successful

Brief Summary

This project successfully implemented an end-to-end Azure data pipeline using Azure Blob Storage and Azure Data Factory. A source CSV file was uploaded to Azure Blob Storage and connected to Azure Data Factory through a Linked Service. The Get Metadata activity validated the source file before processing, and the Copy Data activity transferred the file from the source-container to the destination-container. Pipeline execution was monitored through Azure Data Factory Monitor, and IAM roles were configured to manage resource access securely. The project achieved all required objectives, including metadata validation, successful pipeline execution, and data transfer to the destination location.